Year 13 Maths - Further trigonometry

Similar questions - Compound angle and multiple angle identities

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Prove that $\sin(\theta+45^\circ)\cos(\theta-45^\circ) \equiv \frac{1}{2}(1+\sin 2\theta)$. **[4]**

(b) Hence solve $\sin(\theta+45^\circ)\cos(\theta-45^\circ) = \frac{1}{4}$ for $0^\circ \leqslant \theta < 360^\circ$. **[4]**

2

(a) Use the formulae for $\cos(A+B)$ and the double angle formulae to show that $\cos 3\theta \equiv 4\cos^3\theta - 3\cos\theta$. **[4]**

(b) Hence solve $4\cos^3\theta - 3\cos\theta = \frac{1}{2}$ for $0^\circ \leqslant \theta \leqslant 180^\circ$. **[3]**

(c) Deduce that $\cos 20^\circ$ is a root of $8c^3 - 6c - 1 = 0$. **[2]**

Question 1

(a) $\sin(\theta+45^\circ) = \dfrac{\sin\theta+\cos\theta}{\sqrt{2}}$ and $\cos(\theta-45^\circ) = \dfrac{\cos\theta+\sin\theta}{\sqrt{2}}$. The product is $\dfrac{(\sin\theta+\cos\theta)^2}{2} = \dfrac{1+2\sin\theta\cos\theta}{2} = \frac{1}{2}(1+\sin 2\theta)$

(b) $\frac{1}{2}(1+\sin 2\theta) = \frac{1}{4}$, so $\sin 2\theta = -\frac{1}{2}$. For $0^\circ \leqslant 2\theta < 720^\circ$: $2\theta = 210^\circ, 330^\circ, 570^\circ, 690^\circ$, so $\theta = 105^\circ, 165^\circ, 285^\circ, 345^\circ$

Question 2

(a) $\cos 3\theta = \cos(2\theta+\theta) = \cos 2\theta\cos\theta - \sin 2\theta\sin\theta = (2\cos^2\theta - 1)\cos\theta - 2\sin^2\theta\cos\theta = 2\cos^3\theta - \cos\theta - 2(1-\cos^2\theta)\cos\theta = 4\cos^3\theta - 3\cos\theta$

(b) $\cos 3\theta = \frac{1}{2}$ with $0^\circ \leqslant 3\theta \leqslant 540^\circ$: $3\theta = 60^\circ, 300^\circ, 420^\circ$, so $\theta = 20^\circ, 100^\circ, 140^\circ$

(c) $\theta = 20^\circ$ is a solution, so with $c = \cos 20^\circ$: $4c^3 - 3c = \frac{1}{2}$, giving $8c^3 - 6c - 1 = 0$